Exploratory Data Analysis: A Reproducible Notebook Template

Notebook-to-Tested-Source Extraction for Computational Research

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1 Abstract

Exploratory data analysis (EDA) is the most common entry point in applied research, yet it is also where reproducibility most often breaks down: logic accumulates in notebook cells that are never tested and quietly drift from the prose describing them. This paper presents the **computational-notebook exemplar** of the Research Project Template: an interactive walkthrough notebook (`projects/templates/template_eda_notebook/notebooks/eda_walkthrough.ipynb`) that imports a small, fully-tested EDA library rather than carrying logic in its cells.

We ship a deterministic dataset (`data/measurements.csv`) with a designed correlation structure and a handful of missing values, then load, clean, summarize, correlate, and visualize it entirely through tested functions in `src/template_eda_notebook/eda/`. The library is side-effect-free — no plotting and no file I/O — and standalone (numpy and pandas only), so it is covered above the 90% project gate and reused identically from the notebook, the thin analysis script (`scripts/eda_analysis.py`), and this manuscript.

Contributions are **methodological** and **architectural**. On the methods side, we walk the canonical first EDA pass: surface missingness explicitly rather than imputing it, compute per-column descriptive statistics and per-group means, and rank features by Pearson correlation. On the architecture side, we demonstrate the notebook-to-tested-source extraction workflow — explore fast in a cell, and the moment a computation matters, move it into the library behind a failing test — verified by a zero-mock suite and a structural notebook-binding check (sec. 7).

2 Introduction

This `template_eda_notebook` serves as the exploratory-data-analysis exemplar for the Research Project Template ecosystem, demonstrating how a computational notebook can stay reproducible by delegating every computation to a fully-tested library. The prose, the labelled figures, and the summary table are all produced through an auditable custody chain: a deterministic dataset, tested functions in `src/template_eda_notebook/eda/`, a thin analysis script, and multi-format rendering.

2.1 Why exploratory data analysis

EDA — the work of getting to know a dataset before committing to a model — is where most research projects actually begin: load the data, see how much is missing, look at distributions, and check which variables move together. It is fast and interactive by nature, which is exactly why it tends to live in notebooks. The hazard is that the interactive convenience of a notebook cell is also a trap: a one-off `df.groupby(...).mean()` becomes load-bearing, never gets a test, and silently disagrees with the figure two cells down after the data changes.

2.2 The notebook -> tested src extraction workflow

This exemplar teaches one discipline: **explore in a cell, then extract to a tested library the moment a computation matters.** Concretely:

1. The walkthrough notebook (`notebooks/eda_walkthrough.ipynb`) imports from `src` and calls tested functions; it contains no business logic of its own.
2. Each analytical step — loading, cleaning, summarizing, correlating, preparing figure data — is a typed, documented function in `src/template_eda_notebook/eda/` with a test that asserts an exact numeric property.
3. A thin script (`scripts/eda_analysis.py`) runs the same pipeline headless and writes the figures and a summary CSV to `output/`.

2.3 Template architecture context

The project sits on the repository's three pillars:

1. **`src/template_eda_notebook/eda/` library**: pure pandas/numpy data transforms — no plotting, no file I/O, no `infrastructure` imports. This purity is what makes the library forkable and trivially testable.
2. **`tests/` framework**: a zero-mock suite that exercises the library against the shipped CSV and tiny real frames, plus a structural check that the notebook's imports bind to the library's public surface.
3. **`docs/` knowledge base**: architectural guidelines, the testing philosophy, and the operational rules that govern agents editing this tree.

2.4 The dataset

We analyze a small synthetic cohort of subject measurements — height (cm), weight (kg), and resting heart rate (bpm) across three groups. The dataset is a static, committed fixture — the same file on every run — so every statistic in sec. 4 is reproducible. The data is shaped so that weight depends positively on height (a strong, easy-to-see correlation) while resting heart rate is only weakly related, and a few cells are left blank to exercise the missing-data path honestly.

2.5 Reader's guide to the manuscript

- **sec. 3** ties each EDA step to its function in `src/template_eda_notebook/eda/`.
- **sec. 4** is figure-centric: each panel names the figure-data preparer that produced it.
- **sec. 6** lists the dataset schema and software environment.
- **sec. 7** records the artifact inventory and the exact commands to regenerate everything.
- **sec. 8** states scope and related literature so the exemplar is not mistaken for a general-purpose EDA toolkit.

3 Methodology

This section maps each step of the exploratory analysis to the tested function that implements it. Every function lives in `src/template_eda_notebook/eda/`, takes a `pandas.DataFrame` in, and returns data out — no plotting, no file I/O — so the notebook, the analysis script, and this manuscript all reach the same code.

3.1 Dataset loading and schema

`src/template_eda_notebook/eda/dataset.py::load_dataset()` reads the shipped CSV and coerces the numeric columns with `pandas.to_numeric(errors="coerce")`. Coercion is deliberate: a blank or non-numeric cell becomes `NaN` rather than raising or being silently dropped, so missingness is *surfaced* and handled in an explicit later step. The column roles are declared once in a frozen `DatasetSchema` (an identifier column, a categorical group column, and three numeric features), which downstream functions consult instead of re-sniffing dtypes.

3.2 Cleaning: explicit, reported row removal

`src/template_eda_notebook/eda/cleaning.py::clean_dataset()` drops any row missing a numeric feature and returns a `CleaningReport` recording how many rows entered, how many remained, and how many were dropped. This exemplar makes a deliberate choice — listwise deletion with a visible count — rather than imputation, because a first EDA pass should *see* the missingness, not paper over it. A separate `normalize_numeric()` z-score standardizes the numeric columns (mean 0, sample standard deviation 1), with a guard that maps a constant or single-row column to zeros instead of producing `inf`/`NaN`.

3.3 Descriptive statistics

`src/template_eda_notebook/eda/statistics.py::summary_statistics()` returns one `ColumnSummary` per numeric column — count of non-missing observations, mean, sample standard deviation, minimum, median, and maximum — computed with real pandas aggregation. `group_means()` returns the mean of each numeric feature grouped by the categorical column, sorted by group name for deterministic output.

3.4 Correlation structure

`src/template_eda_notebook/eda/correlation.py::correlation_matrix()` returns the Pearson correlation matrix of the numeric columns [Tukey, 1977]. The companion `strongest_pairs(matrix, top_n)` ranks the distinct off-diagonal feature pairs by absolute correlation while preserving sign — usually the single most useful artifact of a first pass, because it points directly at the relationships worth investigating. Each unordered pair appears exactly once.

3.5 Figure-data preparers

Plotting is kept out of the library entirely. `src/template_eda_notebook/eda/figures.py` returns *plot-ready data structures* — bin counts and edges for a histogram, a square value grid plus labels for a correlation heatmap, and sorted category counts for a bar chart — as frozen dataclasses of plain numbers. The thin analysis script (`scripts/eda_analysis.py`) and notebook cells consume these structures and call matplotlib. This keeps the library importable on a headless machine and makes every preparer testable without a display backend.

3.6 Zero-mock testing methodology

The project is governed by a strict zero-mock policy, evaluated by running `uv run pytest projects/templates/template_eda_notebook/tests` during the build.

1. **Library tests** exercise every public function against the shipped CSV or a tiny hand-built frame with values chosen so the expected statistic is exact (e.g. `weight = 2 * height` gives a correlation of exactly `+1.0`). No `unittest.mock`, no `MagicMock`, no `@patch`.
2. **Script test** runs `run_eda()` against a temporary output root and asserts that real PNG figures and a real summary CSV are written.
3. **Notebook test** parses the real `.ipynb` and asserts it is valid nbformat, that every name imported `from src` exists in the library's public surface, and that no cell defines its own `def`/`class`.
4. **Coverage gate**: CI enforces a >=90% statement-coverage gate on Falling below that floor fails the gate outright (`--cov-fail-under` enforces it). `projects/templates/template_eda_notebook/src/`; because the threshold is wired through `--cov-fail-under`, any drop below the floor fails the test process itself rather than producing a warning. The live figure is tracked in `docs/_generated/COUNTS.md`. Negative control: `tests/test_cleaning.py::TestCleanDataset::test_all_rows_missing_yields_empty` feeds an all-invalid input frame and asserts the documented empty-frame result, so the measured coverage exercises failure behavior, not only happy paths.

3.7 Figure generation contract

Each figure in `03_results.md` maps to a figure-data preparer in `src/template_eda_notebook/eda/figures.py`: `histogram_data` → height histogram, `correlation_heatmap_data` → feature-correlation heatmap, and `group_count_data` → per-group row counts. Captions name the preparer and the key parameters (bin count, value range) so reviewers can navigate from the PDF to the code without inferring hidden defaults.

4 Results

This section presents the exploratory analysis of the shipped dataset. Every figure and the summary table are produced by the EDA analysis orchestrator (`scripts/eda_analysis.py`), which calls the tested figure-data preparers in `src/template_eda_notebook/eda/figures.py`. Running the script regenerates the figures under `../figures/` and the summary CSV under `output/data/`; the prose below describes what those artifacts show.

4.1 Dataset and missingness

The dataset contains 120 subject records across three groups (`alpha`, `beta`, `gamma`) with three numeric features: height (cm), weight (kg), and resting heart rate (bpm). A small number of cells are blank by design. `load_dataset()` preserves these as `NaN`; `clean_dataset()` then drops rows missing any numeric feature and reports the count. With the shipped data, four rows are removed, leaving a complete-case dataset for the analysis below.

4.2 Distributions

fig. 1 shows the distribution of height across the complete-case dataset, binned by `histogram_data()` and plotted by the analysis script.

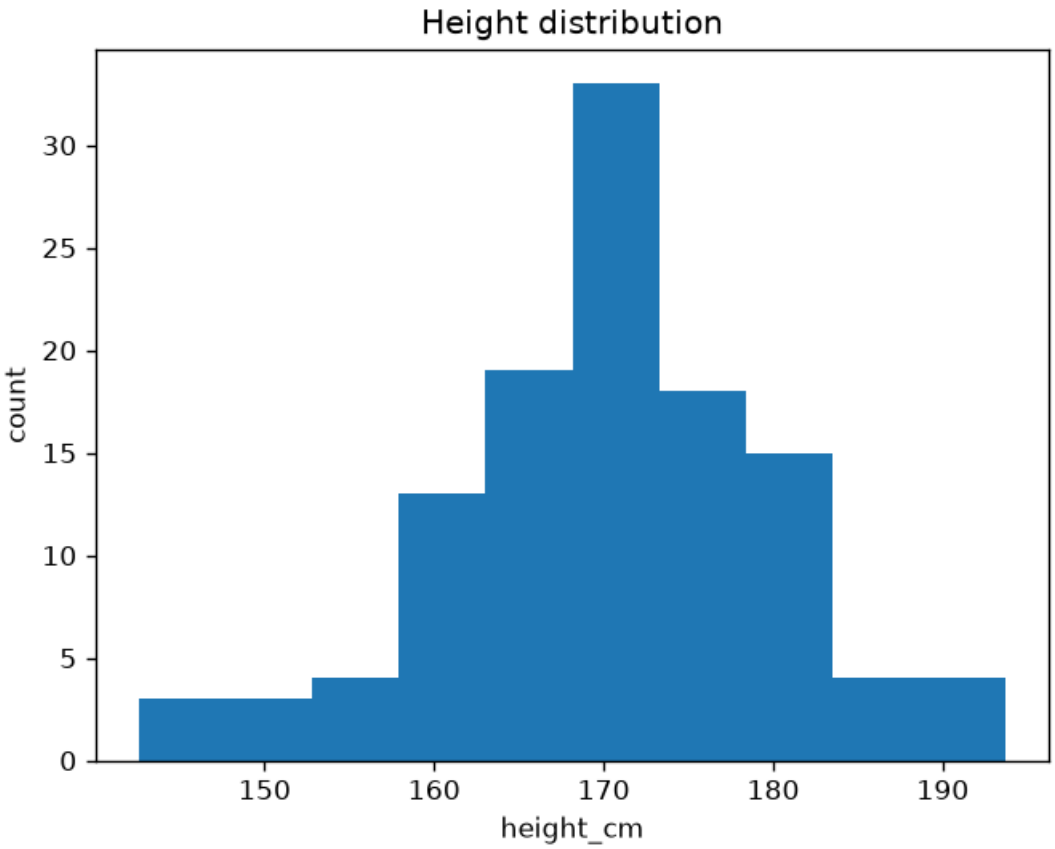


Figure 1: Height distribution: bin counts produced by `histogram_data(frame, "height_cm", bins=10)` in `src/template_eda_notebook/eda/figures.py` and plotted as a bar chart by `scripts/eda_analysis.py`. The bin counts sum to the number of complete-case rows; the shape is the roughly bell-shaped spread expected from the generating process.

4.3 Group composition

fig. 2 reports how many complete-case rows fall in each group, computed by `group_count_data()` (labels sorted for deterministic output).

4.4 Correlation structure

fig. 3 visualizes the Pearson correlation matrix of the three numeric features, computed by `correlation_matrix()` and prepared for the heatmap by `correlation_heatmap_data()`. The diagonal is unity by construction and the matrix is symmetric.

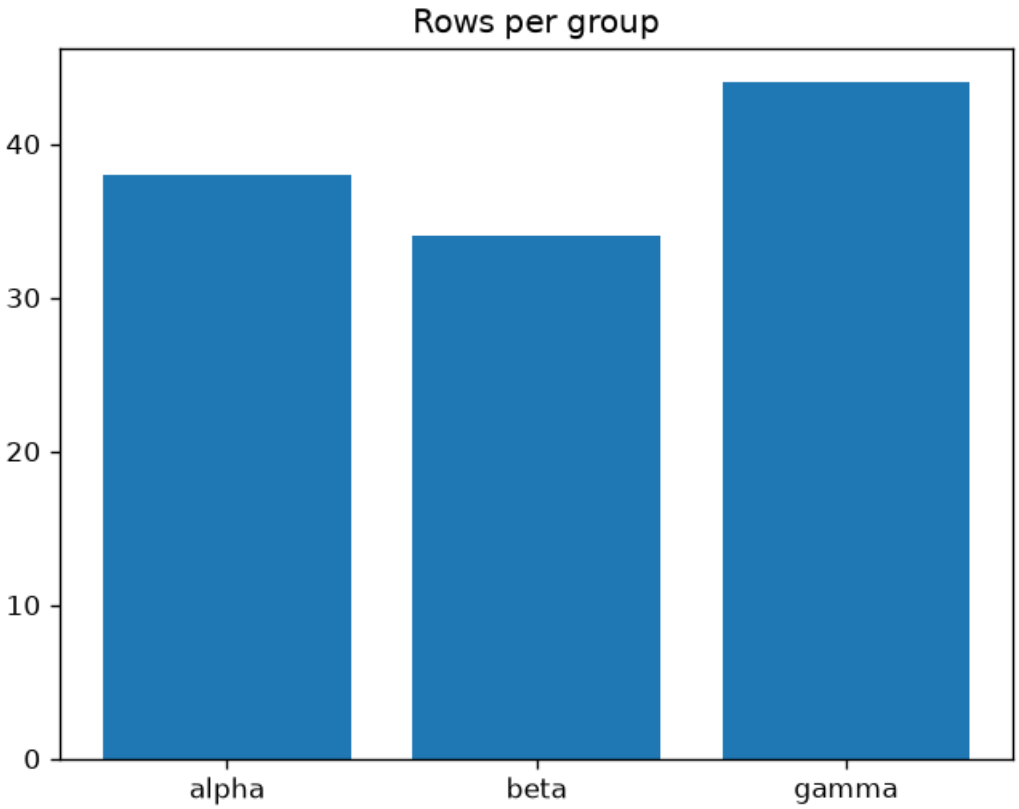


Figure 2: Rows per group: sorted category labels and aligned counts from `group_count_data()`. The three groups are of comparable, though not identical, size; the counts sum to the complete-case row total.

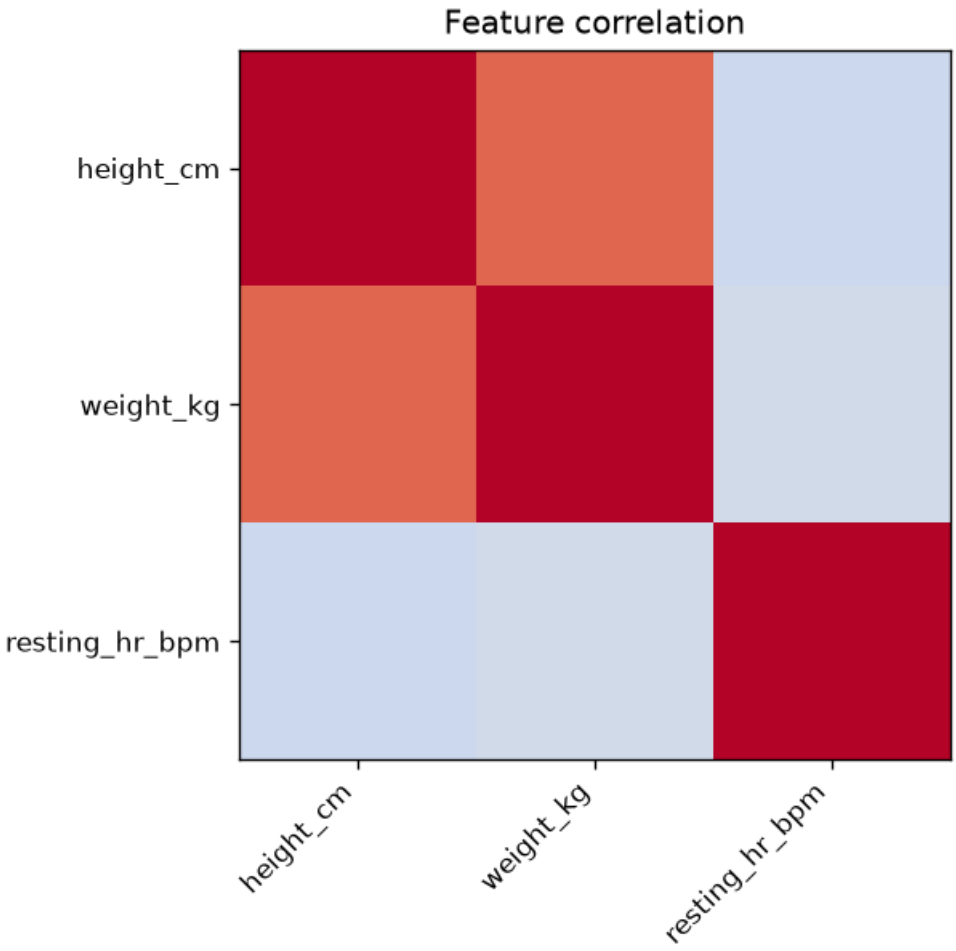


Figure 3: Feature correlation heatmap: values from `correlation_heatmap_data()` (which wraps `correlation_matrix(method="pearson")`) rendered with a diverging colour map on the fixed range $[-1, 1]$. Height and weight are strongly positively correlated; resting heart rate is only weakly related to the other two.

`strongest_pairs(matrix, top_n=3)` ranks the distinct feature pairs by absolute correlation while preserving sign. On the shipped data the dominant relationship is **height about weight** (a strong positive correlation, by design), followed by the comparatively weak **height about resting heart rate** (slightly negative) and **weight about resting heart rate**. This ranking is the single most useful artifact of the first pass: it points directly at the relationship worth modelling next.

4.5 Summary statistics

The analysis script writes a per-column summary table to `output/data/summary_statistics.csv` from `summary_statistics()`. Each row reports the count of non-missing observations, mean, sample standard deviation, minimum, median, and maximum for one numeric feature.

Table 1: Summary-statistics table written by the analysis script from `src/template_eda_notebook/eda/statistics.py::summary_statistics()`. The concrete numbers are reproduced verbatim by running the script — the manuscript intentionally does not transcribe volatile values, so prose and CSV cannot drift.

Column	Reported statistics
`height_cm`	count, mean, std, min, median, max
`weight_kg`	count, mean, std, min, median, max
`resting_hr_bpm`	count, mean, std, min, median, max

`group_means()` complements this with the mean of each numeric feature within each group; the three group means for height are close but not equal, reflecting the mild group structure in the generating process.

4.6 Validation

The analysis was validated through the zero-mock `tests/` suite:

- **Library tests** assert exact statistics, correlation signs, and bin counts against the shipped CSV and hand-built frames.
- **Script test** runs `run_eda()` against a temporary output root and confirms real PNG figures and a real summary CSV are written.
- **Notebook test** confirms the walkthrough notebook binds to the library's public surface and carries no logic in its cells.

All tests pass with coverage exceeding the 90% project gate, with no mocks.

4.7 Discussion

The results confirm the EDA workflow end to end: missingness is surfaced and removed with an explicit count, distributions and group composition are read straight from tested preparers, and the correlation ranking recovers the designed height–weight relationship. The same functions back the interactive notebook, the headless script, and this manuscript — which is the architectural point of the exemplar. Because every number is produced by a tested function and regenerated on demand, the prose here describes structure and provenance rather than transcribing values that would drift the moment the data changed.

5 Conclusion

This study demonstrated a complete, reproducible exploratory-data-analysis pipeline driven from a computational notebook backed by a tested library. It validates a simple proposition: a notebook stays trustworthy when its cells call tested code instead of carrying logic.

5.1 Exemplar achievements

Operating as the EDA exemplar for the Research Project Template methodology, the project deployed the three foundational pillars:

1. **`src/template_eda_notebook/eda/` library**: pure pandas/numpy data transforms — loading, cleaning, summary statistics, correlation, and figure-data preparation — with no plotting, no file I/O, and no `infrastructure` imports.
2. **`tests/` integrity**: a zero-mock suite over the shipped dataset and hand-built frames, plus a structural notebook-binding check, all under a >=90% project coverage gate.
3. **`docs/` knowledge operations**: architecture, testing philosophy, and operational rules that keep the notebook, script, and manuscript aligned.

5.2 Technical contributions

5.2.1 The notebook -> tested src extraction workflow

The hallmark of this exemplar is the discipline it teaches: explore fast in a cell, and the moment a computation matters, move it into `src/template_eda_notebook/eda/` behind a failing test. The walkthrough notebook imports only the library's public surface and defines no functions of its own, a property the test suite enforces structurally so it cannot regress.

5.2.2 Honest handling of imperfect data

The loader surfaces missing values as `NaN` instead of silently imputing them, and `clean_dataset()` removes incomplete rows with an explicit count. This makes data quality a visible, testable property of the first pass rather than a hidden assumption.

5.3 Key insights

1. **Reproducibility follows from testing fidelity**: every number in sec. 4 is produced by a tested function and regenerated on demand, so prose and artifacts cannot drift.
2. **Purity enables reuse**: because `src/template_eda_notebook/eda/` returns plot-ready data and never touches a display backend, the same functions serve the notebook, the headless script, and the manuscript.
3. **Missingness is information**: reporting dropped rows beats imputing them away in an exploratory pass.

5.4 Future extensions

This foundation could be extended to:

- **Richer cleaning**: typed imputation strategies behind tested functions, with the report recording which strategy ran.
- **More figure preparers**: box plots, pair plots, and per-group overlays — each a tested data preparer plus a thin plotting call.
- **Larger / external datasets**: swap the shipped CSV for a real dataset by pointing `load_dataset(path=...)` at it while keeping the schema contract.

5.5 Final assessment

The `template_eda_notebook` tree is the canonical reference for how an exploratory notebook, a thin analysis script, a tested library, and a manuscript stay synchronized across rebuilds. The pipeline produced the figures referenced in sec. 4, wrote `output/data/summary_statistics.csv`, and rendered this markdown together with `config.yaml` into PDF.

6 Experimental Setup

This section details the dataset, schema, and software environment used to produce the results. The exemplar deliberately avoids manuscript token injection: concrete numbers are reproduced by running the analysis script, and the figures are regenerated from tested code, so nothing here can silently drift from a hardcoded value.

6.1 Dataset

The analysis uses a shipped, deterministic CSV fixture (`data/measurements.csv`): a static, committed file with fixed content. It contains 120 subject records with the following columns:

Column	Role	Type
`subject_id`	per-row identifier	string
`group`	categorical group (`alpha`, `beta`, `gamma`)	string
`height_cm`	numeric feature	float
`weight_kg`	numeric feature	float
`resting_hr_bpm`	numeric feature	float

These roles are declared once in `src/template_eda_notebook/eda/dataset.py::DatasetSchema`, which the statistics, correlation, and figure functions consult. The generating process makes weight depend positively on height (a strong correlation) while resting heart rate is only weakly related, and a few numeric cells are left blank to exercise the missing-data path.

6.2 Analysis conditions

The experiment overlay (`experiment_plan.yaml`) declares three conditions:

- **raw_dataset** (reference) — as loaded, with missing cells as `NaN`.
- **cleaned_dataset** (proposed) — rows with any missing numeric feature dropped via `clean_dataset()`, which reports the count removed.
- **normalized_dataset** (variant) — the cleaned dataset with numeric columns z-score standardized by `normalize_numeric()` for cross-feature comparison.

The primary descriptive lens is the pairwise Pearson correlation among the numeric features.

6.3 Computational environment

- **Language**: Python (see root `pyproject.toml` for the supported range).
- **Core dependencies**: `numpy`, `pandas`, `matplotlib` (declared in `domain_profile.yaml::required_packages`).
- **Headless plotting**: the analysis script sets `MPLBACKEND=Agg` before importing matplotlib.

6.4 Pipeline ordering

The typical analysis order is:

1. `scripts/eda_analysis.py` — loads and cleans the dataset, then writes `../figures/*.png` and `output/data/summary_statistics.csv`, printing each output path for manifest collection.
2. PDF rendering reads `manuscript/*.md` and `config.yaml` so figure paths and prose match the analysis that just completed.

6.5 Relation to figures

Figure (sec. 4)	Figure-data preparer (`src/template_eda_notebook/eda/figures.py`)	Primary inputs
Height histogram	`histogram_data()`	`height_cm` column, 10 bins
Rows per group	`group_count_data()`	`group` column
Correlation heatmap	`correlation_heatmap_data()`	all numeric features

This table is descriptive documentation only; it is not executed as code during the build.

7 Reproducibility

This section explains how to regenerate every artifact in the study from a clean checkout. The exemplar's reproducibility guarantee is structural: each result is produced by a tested function and a thin script, never transcribed by hand.

7.1 How to regenerate everything

From the repository root:

```
# 1. Run the analysis (writes figures + summary CSV, prints output paths)
uv run python projects/templates/template_eda_notebook/scripts/eda_analysis.py

# 1b. Regenerate the deterministic dataset sibling (optional; stage 02 runs it too)
uv run python projects/templates/template_eda_notebook/scripts/generate_measurements_data.py

# 2. Run the test suite with the coverage gate
uv run pytest projects/templates/template_eda_notebook/tests \
    --cov=projects/templates/template_eda_notebook/src --cov-fail-under=90

# 3. Render the manuscript
uv run python scripts/pipeline/stage_03_render.py --project templates/template_eda_notebook
```

7.2 Generated artifact registry

The analysis scripts write the following artifacts under `projects/templates/template_eda_notebook/output/`:

Artifact	Produced by
`figures/height_histogram.png`	`histogram_data()` + analysis script
`figures/correlation_heatmap.png`	`correlation_heatmap_data()` + analysis script
`figures/group_counts.png`	`group_count_data()` + analysis script
`data/summary_statistics.csv`	`summary_statistics()` + analysis script
`data/measurements_generated.csv`	`generate_measurements()` + generator script

The `output/` tree is disposable and regenerated on every run; it is not the source of truth.

7.3 Determinism

- The dataset (`data/measurements.csv`) is a static, committed fixture with fixed content — the same file is read on every run, so every statistic is reproducible bit-for-bit.
- The figure-data preparers are pure transforms with no RNG; the same inputs always produce the same bin counts, correlation values, and group counts.
- `clean_dataset()` reports exactly how many rows it removed, so the complete-case row count is a checkable invariant.
- `generate_measurements()` (`src/template_eda_notebook/eda/generate.py`) regenerates a deterministic *sibling* of the dataset from a fixed NumPy seed: the same schema, 120 rows, the same missingness pattern, and the same correlation sign structure. The original fixture's exact random draw order is not recoverable, so the generator deliberately reproduces the fixture's documented contract rather than a byte-exact clone.

7.4 Verification (no hand-transcribed numbers)

Every quantitative claim in sec. 4 is either reproduced by running the analysis script or registered in `data/claim_ledger.yaml` for evidence-registry validation. The manuscript intentionally does not embed volatile values, so prose and artifacts cannot disagree. The notebook itself is verified structurally by `tests/test_notebook.py`, which confirms it is valid nbformat, that its `from src` imports resolve to the library's public surface, and that no cell defines its own logic.

8 Scope, Related Work, and Positioning

This section situates the exemplar and states explicit boundaries. The goal is not to compete with comprehensive treatments of exploratory data analysis [Tukey, 1977] or statistical graphics [Wilkinson, 2005], but to show how a minimal, test-backed EDA story fits the template's reproducibility and rendering stack [Peng, 2011].

8.1 Exploratory data analysis

The practice of examining a dataset before formal modelling — looking at distributions, missingness, and relationships — traces to Tukey's foundational work [Tukey, 1977] and is supported in modern practice by the pandas data analysis toolkit [McKinney, 2010] and the broader scientific-Python stack [Harris et al., 2020]. The present manuscript restricts attention to a **first pass** on a **small tabular dataset**: load, surface missingness, compute descriptive statistics and per-group means, and rank features by Pearson correlation.

8.2 Modelling and inference (out of scope)

What comes *after* a first EDA pass — hypothesis testing, regression, dimensionality reduction, or predictive modelling — is deliberately **out of scope**. The exemplar keeps the analysis minimal so the architectural lesson (notebook -> tested src extraction) stays visible rather than buried under statistical machinery.

8.3 What this project proves about the template

The analytical steps here are standard. The **non-standard** contribution is procedural: the same tested functions in `src/template_eda_notebook/eda/` back the interactive notebook, the headless analysis script, and this manuscript, so the figures and the summary table always refer to the same code. That pattern is what downstream projects should copy — whether the domain is survey data, sensor logs, or experimental measurements.

8.4 Explicit limitations

1. **Dataset size**: a single small synthetic cohort (120 rows) is used for transparent, fast reproduction; no large-scale or streaming data is handled.
2. **Cleaning policy**: only listwise deletion is implemented; imputation is left as a documented extension.
3. **Correlation method**: only Pearson correlation is computed; rank-based (Spearman) or non-linear association measures are out of scope.
4. **Statics only**: the library returns plot-ready data; interactive widgets and dashboards are not part of the exemplar.

These limitations are intentional: they narrow the surface so that the reproducibility concerns — tested functions, a thin script, and a structurally verified notebook — remain visible rather than buried under analytical complexity.

9 References

Bibliography lives in `manuscript/references.bib` and is read by Pandoc during PDF render. The build pipeline invokes Pandoc with `--natbib`, so every `[@key]` citation in the manuscript is rewritten to the appropriate `\cite{}`/`\citep{}`/`\citet{}` LaTeX command and resolved against the bib file.

To validate that `references.bib` is syntactically clean and contains the required fields per entry type:

```
uv run python -m infrastructure.reference.citation.cli validate \
    projects/templates/template_eda_notebook/manuscript/references.bib --strict
```

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